

Windows File Server and Permissions Management Project

Prerequisites

- A functioning Windows Server (ideally Windows Server 2019 or 2022)
- Active Directory Domain Services already configured
- Domain users and groups already created
- Administrative access to the server

Part 1: Installing and Configuring the File Server Role

Step 1: Install File Server Role

1. Log in to your Windows Server with Administrator credentials
2. Open Server Manager (it should launch automatically, or find it in the Start menu)
3. Click on "Manage" in the top-right corner, then select "Add Roles and Features"
4. Click "Next" on the Before You Begin page
5. Select "Role-based or feature-based installation" and click "Next"
6. Select your server from the server pool and click "Next"
7. In the Server Roles list, check "File and Storage Services"
8. Expand "File and Storage Services" and "File and iSCSI Services"
9. Check "File Server" and "File Server Resource Manager"
10. Click "Next" to proceed through the wizard
11. On the Features page, click "Next"
12. Review the information and click "Install"
13. Once installation completes, click "Close"

Step 2: Create a Dedicated Disk Volume for File Shares

For better management, it's recommended to use a separate volume for your file shares:

1. Open Server Manager
2. Click on "File and Storage Services" in the left pane
3. Click on "Disks"
4. Right-click on an available disk and select "New Volume"

5. Follow the New Volume Wizard:

- Select the disk and click "Next"
- Specify volume size and click "Next"
- Assign a drive letter (e.g., E:) and click "Next"
- Select NTFS as the file system, set allocation unit size to default
- Name the volume (e.g., "FileShares") and click "Next"
- Review settings and click "Create"

Part 2: Creating Shared Folders with Proper Permissions

Step 1: Plan Your Folder Structure

Design your folder structure based on organizational needs. Here's an example structure:

E:\FileShares\

```
├── Departments\  
│   ├── HR\  
│   ├── Finance\  
│   ├── Marketing\  
│   └── IT\  
├── Projects\  
│   ├── Project1\  
│   ├── Project2\  
│   └── Project3\  
└── Public\
```

Step 2: Create the Folder Structure

1. Open File Explorer
2. Navigate to your dedicated volume (e.g., E:)
3. Create the folder structure as planned
4. Right-click on the root folder (e.g., "FileShares") and select "Properties"
5. Go to the "Security" tab and click "Advanced"

6. Click "Disable inheritance" and select "Convert inherited permissions into explicit permissions on this object"
7. Modify permissions as needed (keep Administrators with Full Control)

Step 3: Configure NTFS Permissions

For each department folder (example using HR folder):

1. Right-click on the HR folder and select "Properties"
2. Go to the "Security" tab and click "Advanced"
3. Click "Disable inheritance" and select "Convert inherited permissions"
4. Click "Add" to add new permissions
5. Click "Select a principal" and enter "HR-Users" (assuming you've created this security group in AD)
6. In the "Basic permissions" section, check:
 - Read & execute
 - List folder contents
 - Read
 - Write
 - Create folders/append data
 - Create files/write data
7. Click "OK" to apply
8. Add another permission for "HR-Managers" group with "Full control"
9. Remove any unnecessary permissions
10. Click "Apply" and "OK"

Repeat these steps for each department folder with their respective security groups.

Step 4: Create Shared Folders

1. Right-click on the "FileShares" folder and select "Properties"
2. Go to the "Sharing" tab and click "Advanced Sharing"
3. Check "Share this folder"
4. Set the share name as "FileShares"
5. Click "Permissions"

6. In the Share Permissions dialog:
 - Remove "Everyone" if present
 - Add "Domain Users" with "Read" permission
 - Add "Domain Admins" with "Full Control"
 - Click "OK"
7. Click "OK" to close the Advanced Sharing dialog
8. Click "Close" to apply the changes

For individual department shares:

1. Right-click on each department folder (e.g., HR)
2. Select "Properties" > "Sharing" tab > "Advanced Sharing"
3. Check "Share this folder"
4. Name the share (e.g., "HR")
5. Click "Permissions"
6. Configure share permissions:
 - Add the respective department group (e.g., "HR-Users") with "Change" permission
 - Add "HR-Managers" with "Full Control"
 - Add "Domain Admins" with "Full Control"
7. Click "OK" and then "Close" to apply

Part 3: Setting Up Folder Redirection with Group Policy

Step 1: Create a Group Policy Object (GPO)

1. Log in to your domain controller
2. Open Server Manager
3. Click "Tools" > "Group Policy Management"
4. Expand your forest and domain
5. Right-click on the OU where you want to apply folder redirection (e.g., "Users" OU)
6. Select "Create a GPO in this domain, and Link it here"
7. Name it "Folder Redirection Policy" and click "OK"

Step 2: Configure Folder Redirection

1. Right-click the new GPO and select "Edit"
2. In the Group Policy Management Editor, navigate to:
 - User Configuration > Policies > Windows Settings > Folder Redirection
3. Right-click "Documents" and select "Properties"
4. In the Target tab:
 - Setting: "Basic - Redirect everyone's folder to the same location"
 - Target folder location: "Create a folder for each user under the root path"
 - Root Path: \\ServerName\FileShares\Users\%username%\Documents
5. In the Settings tab:
 - Check "Grant the user exclusive rights to Documents"
 - Check "Move the contents of Documents to the new location"
 - Policy Removal: "Redirect the folder back to the local userprofile location"
6. Click "OK" to apply

Repeat these steps for other folders you want to redirect (Desktop, Pictures, etc.).

Part 4: File Classification with File Server Resource Manager

Step 1: Configure File Server Resource Manager

1. Open Server Manager
2. Click "Tools" > "File Server Resource Manager"

Step 2: Create Classification Properties

1. Expand "Classification Management" and click "Classification Properties"
2. Right-click in the right pane and select "Create Property"
3. Configure the property:
 - Property name: "Confidentiality"
 - Description: "Indicates the confidentiality level of the document"
 - Property type: "Single choice list"
 - Values: Add "Public", "Internal", "Confidential", "Restricted"

4. Click "OK" to create the property

Step 3: Create Classification Rules

1. Click on "Classification Rules" in the left pane
2. Right-click in the right pane and select "Create Classification Rule"
3. Configure the rule:
 - Rule name: "Finance Documents Classification"
 - Scope: Browse to "E:\FileShares\Departments\Finance"
 - Classification method: "Folder Classifier"
 - Property: "Confidentiality"
 - Value: "Confidential"
 - Check "Overwrite the existing property value"
4. Click "OK" to create the rule

Repeat to create rules for other departments with appropriate confidentiality levels.

Step 4: Create a File Management Task

1. Expand "File Management Tasks" in the left pane
2. Right-click in the right pane and select "Create File Management Task"
3. Configure the task:
 - Task name: "Confidential Files Expiration"
 - Scope: Select all folders that contain confidential information
 - Conditions: Property "Confidentiality" equals "Confidential" AND File age is more than 365 days
 - Action: "File expiration"
 - Expiration directory: Specify an archive location
4. Configure the schedule to run weekly
5. Click "OK" to create the task

Step 5: Create Storage Reports

1. Click on "Storage Reports Management" in the left pane
2. Right-click in the right pane and select "Schedule a New Report Task"

3. Select reports you want to generate:
 - Duplicate Files
 - Files by Owner
 - Large Files
 - Least Recently Accessed Files
4. Configure report parameters and schedule
5. Specify email delivery options if desired
6. Click "OK" to schedule the report

Part 5: Testing and Verification

Step 1: Test Access Permissions

1. Log in as different users from various departments
2. Verify they can access only their authorized folders
3. Check read/write permissions working as expected
4. Test access from different computers on the network

Step 2: Test Folder Redirection

1. Log in as a test user affected by the folder redirection policy
2. Verify that their Documents folder redirects to the server location
3. Save files to Documents and verify they appear on the server

Step 3: Test File Classification

1. Run the classification rules manually to trigger immediate classification
2. Verify files are properly classified according to rules
3. Check if file management tasks execute properly